as a productivity tool rather than as a substitute for engineering expertise. Its usefulness depends on the problem being addressed and the environment in which it is deployed.

> AI is proving to be a productivity tool rather than a substitute for engineering expertise.

Another common misconception is that AI-generated outputs are inherently reliable. Modern language models can generate convincing code, documentation, and architectural recommendations, but these outputs still require careful validation. AI systems are designed to provide plausible solutions based on the context provided to them, which means they may overlook project-specific constraints or recommend approaches that are unsuitable for enterprise environments. Human review therefore remains essential to verify technical accuracy, evaluate trade-offs, and ensure that AI-generated solutions align with business requirements and organisational policies.

The increasing availability of AI-assisted development tools has also prompted questions about whether software engineering fundamentals remain important. While AI can accelerate implementation, developers still need a strong understanding of algorithms, system architecture, software design principles, and business requirements. Effective AI-assisted development depends on providing clear objectives, accurate context, and well-defined constraints. Without a solid understanding of the underlying system, developers may struggle to determine whether AI-generated recommendations are technically appropriate and aligned with long-term goals.

Rather than making engineering fundamentals less important, AI makes them more relevant to evaluating and directing its output. Developers need to understand system architecture, evaluate technical trade-offs, and recognise when AI-generated recommendations are appropriate. As software systems grow more complex, critical thinking and the ability to provide relevant context become as important as writing code.

The keyboard stays, the job changes

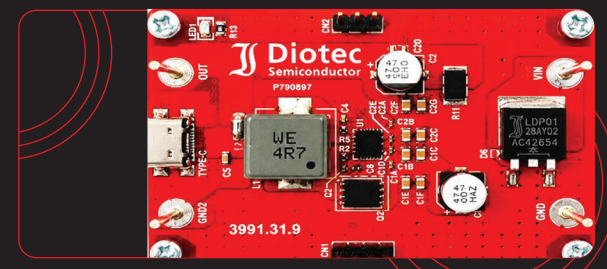
The keyboard is unlikely to disappear from a developer's desk, but its role is changing. As AI handles more routine coding tasks, developers are likely to be valued increasingly for designing systems, solving complex problems, and making informed engineering decisions. Software development is therefore likely to place greater value on developers who combine strong engineering fundamentals with effective use of AI to build reliable and scalable systems. EFY

This article is based on the talk 'AI Is the New Keyboard: Every Developer Must Make The Most Of It' by Mithun Srinivas of Cisco, at AIDEV26. It was transcribed by Ananthu Ashok, Electronics For You.

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